

Orientation and signed Jacobian transfer

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Scope

Metric inverse bounds and the absolute Jacobian formula provide signed transfer. A weak Piola identity shows that the transported sign has zero weak gradient; on a preconnected target the sign is almost everywhere constant and the signed integral is that sign times target volume.

7 direct Cards + 19 reused prerequisites = 26 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [The signed Jacobian integral is a sign times target volume](#)
- Key result: [The transported Jacobian sign has zero weak gradient](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

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Level 0 (9 Cards)

Level 0

Inverse maps on open sets with global metric bounds

Records the maps, their inverse relations, and the quantitative bounds used in the orientation argument.

`MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A positive lower metric bound passes to the derivative](#) · [Weak Piola identity for a compactly supported test field](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Gluing one almost-everywhere constant over a countable cover

Combines local exceptional sets using an inequality between restricted measures.

`MathlibAnnex.aeConstantOn_of_countableCover`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A determinant difference bound in the sup operator norm

Controls determinant variation by replacing one matrix row at a time.

`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Determinant integrals under strong \$L^n\$ convergence](#)

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Level 0

Mollification by a normalized smooth kernel

Defines convolution with a nonnegative smooth kernel of integral one.

`MathlibAnnex.Mollification.mollify`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Strong local convergence of mollified derivatives](#)

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Level 0

Divergence as the trace of a derivative

Defines divergence without choosing coordinates.

`MathlibAnnex.divergence`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Vanishing weak gradient tested by divergence](#) · [Weak Piola identity for a compactly supported test field](#)

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Level 0

A cofactor row by row replacement

Fixes the cofactor convention used in the divergence and determinant identities.

`MathlibAnnex.Piola.cofactorRow`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The divergence-free cofactor identity](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Zero integral of a compactly supported divergence

Makes all faces in a box divergence formula vanish by placing the support strictly inside the box.

`MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Identifying constants on an open overlap

Uses positive measure to find a point where both almost-everywhere equalities hold.

`MathlibAnnex.aeConstants_eq_of_open_overlap`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Differentiating a local mollification through its kernel

Writes a directional derivative without differentiating the locally integrable input.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (6 Cards)

Level 1

A positive lower metric bound passes to the derivative

Preserves the same lower constant in the difference-quotient limit.

`MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound`

Immediate Card prerequisites: [Inverse maps on open sets with global metric bounds](#)

Used by in this scope: [Signed transfer from the absolute Jacobian formula](#)

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Level 1

Strong local convergence of mollified derivatives

Approximates the derivative in local L^n , where n is the dimension of the domain.

`MathlibAnnex.Mollification.tendsto_eLpNorm_fderiv_mollify_sub`

Immediate Card prerequisites: [Mollification by a normalized smooth kernel](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

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Level 1

The divergence-free cofactor identity

Cancels the Hessian terms that arise when differentiating a cofactor row.

`MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero`

Immediate Card prerequisites: [A cofactor row by row replacement](#)

Used by in this scope: [A component flux gives a determinant difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Vanishing weak gradient tested by divergence

Expresses a weak equation through compactly supported continuously differentiable vector fields.

`MathlibAnnex.WeakDivergenceZero`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#)

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#) · [The transported Jacobian sign has zero weak gradient](#)

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Level 1

From local to global almost-everywhere constancy

Separates agreement of constants by connectedness from countable measure-theoretic gluing.

`MathlibAnnex.exists_aeConstantOn_of_local`

Immediate Card prerequisites: [Gluing one almost-everywhere constant over a countable cover](#) · [Identifying constants on an open overlap](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

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Level 1

Determinant integrals under strong L^n convergence

Uses Hölder's inequality to turn strong L^n convergence of matrix fields into convergence of their determinant integrals.

`MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn`

Immediate Card prerequisites: [A determinant difference bound in the sup operator norm](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

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Level 2 (3 Cards)

Level 2

Signed transfer from the absolute Jacobian formula

Transfers an integrable function without first assuming that the Jacobian sign is constant.

`MathlibAnnex.BilipschitzOrientation.signed_area_transfer`

Immediate Card prerequisites: [A positive lower metric bound passes to the derivative](#)

Used by in this scope: [The transported Jacobian sign has zero weak gradient](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A component flux gives a determinant difference

Expresses the change from one output-coordinate perturbation as a divergence.

`MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`

Immediate Card prerequisites: [The divergence-free cofactor identity](#)

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

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Level 2

Zero weak gradient gives zero derivatives of local mollifications

Substitutes a translated smooth kernel into the weak equation and tracks the reflection sign.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Differentiating a local mollification through its kernel](#)

Used by in this scope: [Local almost-everywhere constancy from the weak equation](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (2 Cards)

Level 3

Smooth compact perturbations preserve the determinant integral difference

Builds a finite telescope from single-output perturbations with compactly supported fluxes.

```
MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff
```

Immediate Card prerequisites: [A component flux gives a determinant difference](#) · [Zero integral of a compactly supported divergence](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

Local almost-everywhere constancy from the weak equation

Turns constant mollifications into one almost-everywhere constant by choosing a common convergence point.

```
MathlibAnnex.WeakDivergenceZero.nonempty_localAEConstantAt
```

Immediate Card prerequisites: [Zero weak gradient gives zero derivatives of local mollifications](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (2 Cards)

Level 4

Global almost-everywhere constancy on a preconnected domain

Combines local weak-gradient rigidity with topological and countable gluing.

```
MathlibAnnex.WeakDivergenceZero.exists_aeConstantOn
```

Immediate Card prerequisites: [Local almost-everywhere constancy from the weak equation](#) · [From local to global almost-everywhere constancy](#)

Used by in this scope: [Almost-everywhere constancy of the sign on a preconnected target](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4

Maximal-minor integral differences for Lipschitz perturbations

Passes the smooth compact-perturbation identity to Lipschitz maps through local strong convergence.

```
MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith
```

Immediate Card prerequisites: [Determinant integrals under strong \$L^n\$ convergence](#) · [Strong local convergence of mollified derivatives](#) · [Smooth compact perturbations preserve the determinant integral difference](#)

Used by in this scope: [Weak Piola identity for a compactly supported test field](#)

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Level 5 (1 Card)

Level 5

Weak Piola identity for a compactly supported test field

Uses coordinate perturbations to show that the signed pullback of a test-field divergence has zero integral.

`MathlibAnnex.BilipschitzOrientation.weak_piola`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#) · [Maximal-minor integral differences for Lipschitz perturbations](#) · [Inverse maps on open sets with global metric bounds](#)

Used by in this scope: [The transported Jacobian sign has zero weak gradient](#)

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Level 6 (1 Card)

Level 6

The transported Jacobian sign has zero weak gradient

Uses signed transfer and weak Piola with the same test field to obtain the weak equation on the target.

`MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign`

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Signed transfer from the absolute Jacobian formula](#) · [Weak Piola identity for a compactly supported test field](#)

Used by in this scope: [Almost-everywhere constancy of the sign on a preconnected target](#)

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Level 7 (1 Card)

Level 7

Almost-everywhere constancy of the sign on a preconnected target

Obtains one constant from the weak equation and preconnectedness of the target.

`MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const`

Immediate Card prerequisites: [Global almost-everywhere constancy on a preconnected domain](#) · [The transported Jacobian sign has zero weak gradient](#)

Used by in this scope: [The signed Jacobian integral is a sign times target volume](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8 (1 Card)

Level 8

The signed Jacobian integral is a sign times target volume

Uses preconnectedness to obtain a constant, positive measure to identify its sign, and finite measure to integrate it.

`MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume`

Immediate Card prerequisites: [Almost-everywhere constancy of the sign on a preconnected target](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)